

MACON-CHARNAY AP, France

WMO: 073850

Lat: 46.2967N

Lon: 4.7986E

Elev: 222

StdP: 98.69

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-5.7	-3.9	-9.3	1.7	-4.1	-7.1	2.1	-2.2	10.2	5.6	8.9	3.5	2.5	320	0.466

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.6	32.5	20.7	30.5	20.2	28.6	19.5	21.8	29.7	20.9	28.4	20.2	27.0	3.4	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.2	14.3	24.8	18.4	13.7	24.0	17.7	13.0	23.2	64.4	29.9	61.4	28.4	58.7	27.1	26.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.3	7.1	6.1	-8.5	35.1	3.0	2.2	-10.7	36.7	-12.4	37.9	-14.1	39.1	-16.3	40.7		
(5)			-8.9	23.4	2.9	1.0	-11.0	24.1	-12.7	24.6	-14.4	25.2	-16.5	25.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	12.2	3.3	4.6	8.3	11.6	15.5	19.5	21.4	20.8	16.8	12.8	7.3	3.9		
	DBStd	7.33	4.02	3.82	3.58	3.46	3.40	3.50	3.22	3.19	3.06	3.65	3.79	4.12		
	HDD10.0	764	208	153	75	23	2	0	0	0	0	16	97	191		
	HDD18.3	2525	465	384	311	203	99	28	8	11	66	173	330	447		
	CDD10.0	1568	1	2	22	69	173	285	354	336	204	103	16	2		
	CDD18.3	288	0	0	0	1	13	63	104	88	19	2	0	0		
	CDH23.3	2558	0	0	0	14	124	556	928	770	155	11	0	0		
	CDH26.7	865	0	0	0	1	20	180	343	294	27	0	0	0		
(14) Wind	WSAvg	2.6	2.6	2.9	3.1	3.0	2.8	2.7	2.6	2.3	2.4	2.3	2.5	2.5		
(15) Precipitation	PrecAvg	874	63	56	50	71	80	68	76	75	68	93	104	70		
	PrecMax	1023	134	107	143	175	143	144	136	200	203	196	265	124		
	PrecMin	622	17	6	9	15	16	12	18	14	7	17	42	17		
	PrecStd	108	29	27	29	40	31	33	31	37	47	38	53	28		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.0	15.7	20.8	25.4	29.2	34.2	35.1	36.1	29.7	24.8	18.6	15.0		
		MCWB	10.6	10.1	12.6	15.9	18.4	20.9	20.5	21.0	19.1	17.2	13.9	11.2		
	2%	DB	11.8	13.5	18.6	22.6	26.5	30.9	32.4	32.2	27.1	22.5	16.3	12.6		
		MCWB	9.3	9.2	11.5	14.1	17.2	20.4	20.6	20.7	18.5	16.4	12.6	10.0		
	5%	DB	10.3	11.9	16.4	20.4	24.5	28.7	30.4	30.0	25.1	20.4	14.5	11.0		
		MCWB	8.3	8.4	10.8	13.1	16.5	19.6	20.2	20.2	17.9	15.7	11.8	9.2		
	10%	DB	9.0	10.1	14.4	18.3	22.4	26.7	28.4	27.6	23.0	18.4	12.8	9.6		
		MCWB	7.4	7.5	9.8	12.1	15.6	18.7	19.6	19.5	16.8	14.7	10.8	8.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.2	10.8	13.6	16.7	20.0	22.7	22.6	22.8	20.8	18.4	14.5	11.6		
		MCDB	13.2	14.3	18.9	23.9	26.8	30.9	31.1	31.1	26.8	22.5	17.5	13.8		
	2%	WB	9.5	9.7	12.2	15.0	18.4	21.3	21.7	21.6	19.5	17.2	13.2	10.4		
		MCDB	11.3	12.8	17.0	20.9	24.6	28.5	29.9	29.6	25.4	21.0	15.6	12.3		
	5%	WB	8.5	8.7	11.2	13.7	17.2	20.3	20.8	20.7	18.4	16.2	12.1	9.3		
		MCDB	10.1	11.3	15.6	19.1	22.7	27.1	28.6	27.9	23.5	19.4	14.2	10.8		
	10%	WB	7.5	7.7	10.2	12.6	16.1	19.3	20.0	19.9	17.4	15.1	11.0	8.3		
		MCDB	8.8	9.9	13.9	17.3	21.2	25.4	26.9	26.4	21.8	18.0	12.8	9.5		
(35) Mean Daily Temperature Range	5% DB	MDBR	5.3	6.9	9.1	9.9	9.8	10.4	10.6	10.8	10.2	8.4	6.1	5.1		
		MCDBR	7.1	10.1	12.8	13.7	13.4	14.0	14.0	15.2	13.5	11.4	8.4	7.1		
	5% WB	MCWBR	5.1	6.5	7.1	6.9	6.1	5.8	5.3	5.9	6.3	6.2	5.5	5.2		
		MCWBR	6.7	9.0	11.4	12.3	11.9	12.5	12.4	12.9	11.7	10.3	7.6	6.5		
(40) Clear-Sky Solar Irradiance	taub	taub	0.312	0.331	0.359	0.394	0.399	0.404	0.401	0.395	0.374	0.361	0.329	0.308		
		taud	2.464	2.408	2.346	2.250	2.288	2.309	2.315	2.341	2.380	2.415	2.460	2.468		
	Ebn at Noon	Ebn at Noon	789	841	864	861	868	863	860	851	838	793	757	749		
		Edh at Noon	68	87	106	128	128	126	124	116	102	86	68	61		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	1.17	2.01	3.30	4.49	5.24	6.21	5.99	5.18	3.98	2.47	1.33	0.96		
	RadStd	RadStd	0.14	0.31	0.43	0.61	0.53	0.54	0.51	0.41	0.31	0.22	0.13	0.15		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (40 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page